Not All Pairs Need All Correspondences: Certified Adaptive Budgets for Robust Point Cloud Registration

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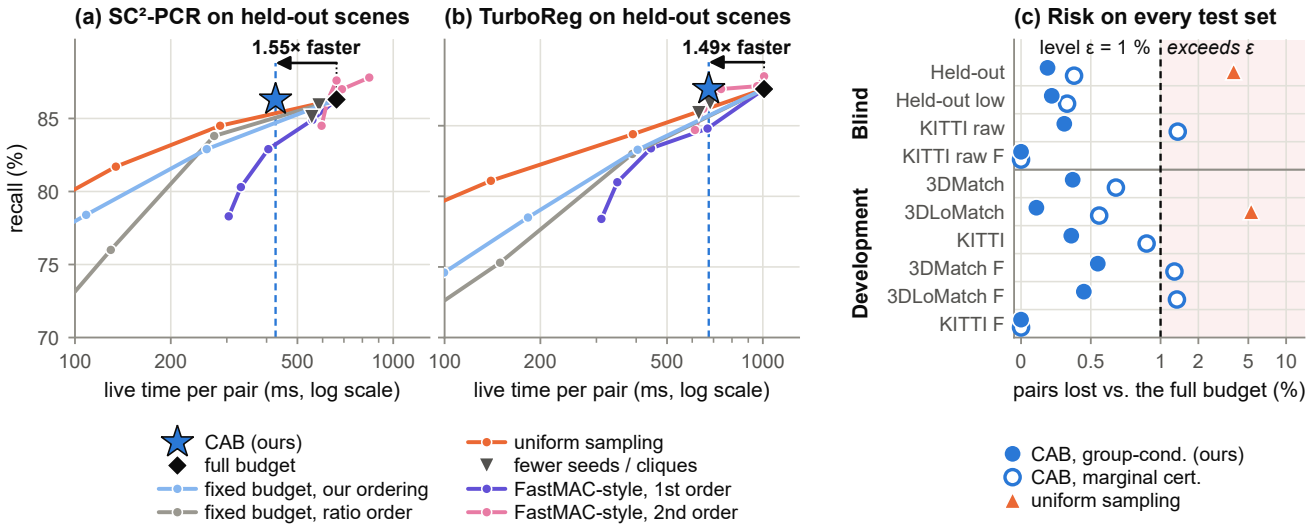


Figure 1. **Cheaper is easy; safe is not.** (a, b) Recall against live time per pair on the blind held-out scenes (overlap above 30 %) for every cheaper way of running the same estimator (Tab. 3 lists every configuration): fixed budgets of the most consistent (light blue) or most distinctive (grey) correspondences, uniform sampling (orange), FastMAC-style spectral sampling on the first- (violet) or second-order graph (pink), and fewer seeds or cliques (triangles). CAB (blue star) adapts the budget to each pair; the dashed line marks its time and the arrow its speed-up over the full budget in this timing run (Tab. 1 reports the main timing run). (c) Risk: the share of pairs registered by the full estimator but lost, on every blind test set and development benchmark (F: FCGF correspondences), for each method at its worst over the two estimators and the two orderings. Only CAB with group-conditional certificates (filled blue) stays below the certified level of 1 % everywhere; marginal certificates (open) and uniform sampling of 2,000 correspondences exceed it on some test sets.

Abstract

The most accurate robust estimators for point cloud registration build a compatibility graph over all putative correspondences and pay this cost on every pair, although most pairs can be registered from a small, well-chosen subset. Making them cheaper is easy; making them cheaper safely *is not: every fixed shortcut, such as fewer correspondences, sampling or fewer seeds, is tuned to one benchmark's mix of easy and hard pairs and silently loses pairs when the mix changes. We present certified adaptive budgets (CAB), a drop-in wrapper that runs an unchanged estimator on growing budgets of mutually consistent correspondences and stops as soon as a learned controller is confident. The stopping rule is certified per difficulty group with Learn-then-Test, so that with high probability at most 1 % of the pairs registered by the full estimator are lost, for any mix of difficulties. We evaluate it under a pre-registered protocol on blind test sets, held-out indoor scenes and KITTI raw drives never used during development, with hand-crafted and learned descriptors. Around SC²-PCR and TurboReg, CAB is 1.56× and 1.49× faster on the held-out scenes with statistically non-inferior recall, and its realized risk stays within the certified level on all 40 test conditions, whereas marginal certificates and uniform sampling exceed it. Where nearly every pair needs the full budget, at low overlap and on the raw drives, CAB declines to cut instead of losing pairs, at a cost of at most 10 % in*

time.

1. Introduction

Correspondence-based registration aligns two scans from putative point correspondences proposed by local descriptors. The robust estimator that turns them into a rigid motion is where most of the accuracy comes from, and increasingly where most of the time goes. The most accurate estimators today, such as SC^2-PCR [6], MAC [33] and TurboReg [30], build a compatibility graph over all N correspondences, at a cost that grows quadratically or cubically with N. They spend this cost uniformly: every configuration fixes N for every pair, and evaluations sweep it from 250 to 5,000 [15, 23].

Yet pairs differ enormously in difficulty. Run on only 250 correspondences chosen by our ordering (Sec. 3.2), SC^2-PCR already registers 68.8 % of the pairs of our held-out indoor scenes in 72 ms, against 85.2 % in 2,690 ms with all 4,000. Other pairs need every correspondence, and some cannot be registered at all.

Cheaper is easy; safe is not. The obvious remedy is a cheaper fixed configuration, and Fig. 1(a) shows that one is easily found: uniform sampling of 2,000 correspondences runs SC^2-PCR in 286 instead of 664 ms on the held-out scenes. But a fixed shortcut is tuned to one benchmark's mix of easy and hard pairs, and nothing tells its user what it loses: 3.35 % of the pairs the full estimator registers on the held-out scenes and 4.83 % on low-overlap 3DLoMatch (Fig. 1(c)). A speed-up that drops an unknown share of the pairs the trusted estimator would have registered is not one that practitioners can adopt. What they need is a *contract*: a bound on what is lost relative to the estimator they trust, which survives a change of data.

Certified adaptive budgets. We provide such a contract with *certified adaptive budgets* (CAB), a drop-in wrapper around an unchanged estimator (Algorithm 1). CAB runs the estimator on a ladder of growing budgets and lets a learned controller stop after any stage. Two ingredients make this both fast and safe. First, the budgets are filled by a *compatibility-guided ordering* that puts mutually consistent correspondences first, scoring every correspondence against a small set of inlier-rich anchors at a cost far below that of the estimator's own graph (Fig. 2). Second, the controller's operating point is certified with Learn-then-Test [2]: with probability $1-\delta$, at most a fraction ε of the pairs that the full estimator registers are lost, and this holds *separately for every difficulty group*, hence for any mix of easy and hard pairs. The second point matters in practice: marginal certificates, valid only for the mix of difficulties they were calibrated on, exceed their level when the test mix differs (up to 1.37 % on our test sets), while difficulty-conditional ones hold on every test set.

Our contributions are:

- A drop-in wrapper that adapts the correspondence budget of a graph-based estimator to each pair, with a certified bound on the pairs lost against the full estimator that is valid for any mixture of difficulties (Sec. 3).
- A two-pass anchor ordering that fills small budgets with consistent correspondences at a fraction of the cost of the full graph, and an analysis of which pairs it can rescue (Secs. 3.2 and 4.5).
- Evidence under a pre-registered protocol, with blind test sets (held-out 3DMatch training scenes and KITTI raw drives outside the odometry benchmark), hand-crafted (FPFH) and learned (FCGF) descriptors, live timing and paired tests, that the certified wrapper stays within its risk level on every test set, while marginal certificates and fixed shortcuts (fewer seeds or cliques, uniform or FastMAC-style sampling, fixed budgets) do not (Secs. 4.2 to 4.4).

On the blind held-out scenes CAB is 1.56× faster for SC^2-PCR and 1.49× for TurboReg, and on the official 3DMatch benchmark 1.58× and 1.42×. Where the data leave nothing to save, at low overlap and on outdoor raw drives with FPFH, it declines to cut rather than losing pairs, and we show that the remaining time is spent on pairs that no budget can register or that need every correspondence; where its controller cannot tell success from failure, it runs the full budget.

2. Related work

Graph-based robust estimators. Spatial compatibility between correspondence pairs [3] is the backbone of today's most accurate robust estimators. SC^2-PCR [6] scores pairs by second-order compatibility and grows consensus sets from spectral seeds, MAC [33] enumerates maximal cliques, and TurboReg [30] replaces clique enumeration by pivot-guided search of 3-cliques; TEASER [31] and KISS-Matcher [18] prune outliers with certifiable or graph-theoretic tests. All of them build a graph over the N putative correspondences, so their cost is set by N, which each configuration fixes for every pair. FastMAC [34] reduces the number of correspondences passed to MAC by stochastic spectral sampling, but computes its sampling signal on the full second-order graph and uses one sampling ratio for all pairs; TurboReg notes that downsampling correspondences speeds up clique search at a cost in accuracy [30]. These works are complementary to ours: CAB leaves the estimator unchanged, wraps any of them, and decides per pair how many correspondences it needs, with a bound on what this costs in accuracy; its ordering uses the same pairwise test, but only against a small anchor set.

Sampling and stopping in RANSAC. RANSAC [11] adapts its number of hypotheses to an estimate of the inlier

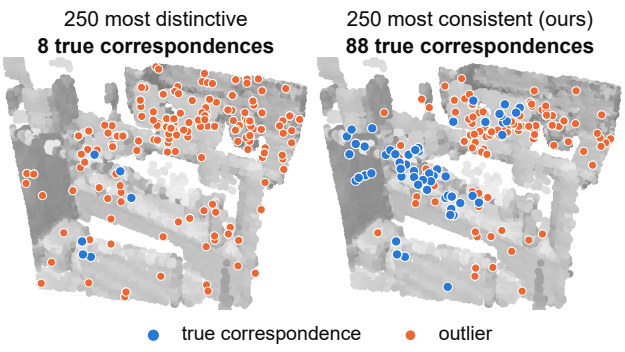


Figure 2. **What the cheap budget contains.** The 250 correspondences given to the estimator on a 3DMatch test pair, drawn at their reference endpoints on the reference fragment, seen from slightly above its sensor. Left: the 250 most distinctive (Lowe ratio). Right: the 250 most consistent with inlier-rich anchors (ours). Blue: true correspondences; orange: outliers.

ratio. PROSAC [8] draws from quality-ordered correspondences, sequential tests [19] and preemption [20] stop evaluating poor hypotheses early, MAGSAC++ [5] and SupeRANSAC [4] improve scoring and sampling, BANSAC [21] adapts sampling online, and Schönberger *et al*. [26] show that the classical stopping criterion is miscalibrated. These rules stop when a model seems good enough, but none bounds the accuracy lost relative to a reference estimator; we compare a certified adaptive pose search with RANSAC in Supp. F.

Adaptive computation with guarantees. Early exits and adaptive computation time [13, 28] let networks spend less on easy inputs, and CALM [27] and Fast-yet-Safe [16] certify early exits with Learn-then-Test [2], a distribution-free risk-control procedure. We bring this guarantee to robust geometric estimators, where the reference is the full estimator rather than a network's last layer, and show that its marginal form breaks under the shifts in difficulty that separate registration benchmarks. In registration, DIPR [1] and BUFFER-X-Lite [17] stop iterations or feature scales early, with hand-set thresholds and without a bound on the pairs lost. Monitoring anytime algorithms by dynamic programming [14] and the value of computation [24] motivate our Markov controller, and our difficulty-conditional certificates follow the group-conditional (Mondrian) guarantees of conformal prediction [29].

3. Certified adaptive budgets

3.1. Problem setup

A pair of scans comes with N putative correspondences $\mathcal{C} = \{(\mathbf{p}_i, \mathbf{q}_i)\}_{i=1}^{N}$ from descriptor matching, sorted by distinctiveness (Lowe ratio). A robust estimator E maps any subset of $\mathcal{C}$ to a rigid transformation; the standard configuration runs it once on all N correspondences, the *full budget*, which is our *reference*. CAB runs E along a ladder of budgets $N_1 < \cdots < N_K = N$: stage k returns $\hat{T}_k = E(\mathcal{C}_{\pi,N_k})$, the estimate from the first N_k correspondences of an ordering π, at cumulative cost C_k. Let $s_k \in \{0, 1\}$ indicate that $\hat{T}_k$ is correct. A stopping policy observes signals z_k after each stage and stops at τ, returning $\hat{T}_\tau$. We want the cheapest policy whose *risk*, the fraction of pairs the reference registers but the policy does not, is at most ε:

$$\min_{\tau} \mathbb{E}[C_\tau] \quad \text{s.t.} \quad R(\tau) = \Pr\left(s_K = 1,\ s_\tau = 0\right) \le \varepsilon. \tag{1}$$

R upper-bounds the recall lost against the standard estimator. Measuring the risk against the reference rather than against ground truth isolates the price of adaptivity: CAB does not claim to fix the estimator's failures, only to bound how many of its successes it gives up, which is what a practitioner who has chosen an estimator cares about. Ground truth is needed only on calibration pairs, never at test time. Two things decide how cheap a certified policy can be: how often the early stages already succeed, which depends on the ordering π (Sec. 3.2), and how well the controller recognises those successes (Secs. 3.3 and 3.4).

3.2. Compatibility-guided ordering

Distinctiveness is a poor filter for a small budget: on the 3DMatch validation scenes, only 27.8 of the 250 most distinctive correspondences are inliers on average. Inliers, however, are mutually consistent. Rigidity implies $\big|\|\mathbf{p}_i - \mathbf{p}_j\| - \|\mathbf{q}_i - \mathbf{q}_j\|\big| < d$ for two inliers i, j, where d is the estimator's compatibility threshold; let C_{ij} indicate this test. Scoring all pairs (i, j) costs $O(N^2)$, as much as the estimator itself. We instead score every correspondence against a small set A of anchors with second-order support,

$$s_i = \sum_{a \in A} C_{ia} \left(C\, C_{AA}\right)_{ia}, \tag{2}$$

the number of anchor triangles that i closes, and fill each budget with the highest-scoring correspondences (ties keep the ratio order). A budget that covers all correspondences keeps the standard input, so the full stage is exactly the reference.

Anchors decide. The obvious anchors, the $m = 100$ most distinctive correspondences, fail exactly where help is needed: in hard pairs they are nearly all outliers (12.5 inliers on average and none in 15 % of validation pairs), and outliers that happen to agree with each other are promoted. We therefore select anchors in a first pass: among the $P = 500$ most distinctive correspondences, the $m = 100$ with the highest first-order degree $\sum_{j \le P} C_{ij}$ become anchors, which raises the share of inliers among anchors to 35.6 %. The two passes cost $O(P^2 + Nm^2)$ instead of the $O(N^2)$ to $O(N^3)$ of the estimators' graphs, 7.2 ms per pair on one CPU core (Fig. 2).

Algorithm 1 CAB at test time

Require: correspondences $\mathcal{C}$ (ratio order), estimator E, ladder $N_1 < \cdots < N_K$, models p_{now}, group edges, certified thresholds λ_g

$\pi \leftarrow$ two-pass compatibility order of $\mathcal{C}$ (Eq. (2)) ▷ skipped if $N_1 \geq |\mathcal{C}|$
for $k = 1, \ldots, K$ **do**
  $\hat{T}_k \leftarrow E(\text{first } N_k \text{ of } \pi(\mathcal{C}))$ ▷ all of $\mathcal{C}$, in ratio order, if $N_k \geq |\mathcal{C}|$
  $z_k \leftarrow$ signals of $\hat{T}_k$ on $\mathcal{C}$ and change from $\hat{T}_{k-1}$
  if $k = 1$ **then** $g \leftarrow$ difficulty group of z_1
  end if
  if $p_{\text{now}}(z_k) \geq \lambda_g$ **or** $k = K$ **then return** $\hat{T}_k$
  end if
end for

3.3. Stopping controller

After each stage the controller observes estimator-agnostic signals: the estimator's inlier count; the fraction of *all* N correspondences, and of the 250 most distinctive, explained by $\hat{T}_k$; the change of $\hat{T}_k$ since the previous stage; the gain in explained correspondences; the stage index, $\log N$ and the median Lowe ratio of the budget. Two logistic models fitted on training pairs predict $p_{\text{now}} = \Pr(s_k{=}1 \mid z_k)$ and the value of continuing, $p_{\text{fin}} = \Pr(s_K{=}1 \mid z_k)$. We consider three policy families: a *threshold* rule that stops when $p_{\text{now}} \geq \lambda$; *accept-or-give-up*, which also stops when $p_{\text{fin}} < \gamma$; and a *Markov* controller, a finite-horizon decision process over (stage, quintile of p_{now}, tercile of p_{fin}) whose transitions and costs are estimated on training trajectories and which is solved by backward induction [22] for the reward $s - \lambda C$. The ladder itself is a design choice: candidates are increasing subsets of $\{250, 500, 1000, 2000, 4000\}$, including ladders that end below the full budget.

3.4. Certification

Every policy family is indexed by a parameter ordered from conservative to aggressive. On n held-out calibration pairs, Learn-then-Test [2] tests $H_\lambda : R(\lambda) > \varepsilon$ in this fixed order with exact binomial p-values and keeps the most aggressive λ before the first non-rejection; then $\Pr\left(R(\hat{\lambda}) \leq \varepsilon\right) \geq 1 - \delta$ for test pairs exchangeable with the calibration pairs. We use $\varepsilon = 1\,\%$ and $\delta = 0.1$. A certificate needs data: even with no loss among n calibration pairs, the test rejects only if $(1-\varepsilon)^n \leq \delta$, *i.e.* for $n \geq 230$ here.

Difficulty-conditional certificates. Benchmark splits differ mostly in their mix of easy and hard pairs, which breaks exchangeability between calibration and test pairs. We therefore partition pairs into G groups by an observable after the first stage, the number of explained correspondences, at its terciles on the fitting pairs, and certify one threshold λ_g per group, each on that group's calibration pairs at level δ/G (which needs at least 339 pairs per group for $G = 3$).

Proposition 1. *If, within every group g, test pairs are exchangeable with that group's calibration pairs, then with probability at least $1 - \delta$ the policy with thresholds $(\hat{\lambda}_g)_g$ has $R \leq \varepsilon$ on every test distribution, whatever its group proportions.*

Proof sketch. By a union bound over the G group-wise guarantees, $R_g(\hat{\lambda}_g) \leq \varepsilon$ holds for all g simultaneously with probability $\geq 1-\delta$; then $R = \sum_g w_g R_g \leq \varepsilon$ for any proportions w_g (full proof in Supp. A). □

A marginal certificate only covers test sets with the calibration mix of difficulties; the conditional one covers any mix, at the price of a smaller δ per group and hence more calibration pairs.

Choose first, then certify. The guarantee covers one sequence of tests fixed in advance, so the ladder and the policy family are chosen before any calibration pair is used. On the fitting pairs, scored out of fold by controllers fitted on the other folds, every candidate gets the most aggressive parameters whose losses stay within what the certificate can accept for the calibration set's size (none in any group, with our pool sizes), and we keep the cheapest candidate; only this one is certified, and the result is evaluated once on test. Certifying several candidates on the same calibration pairs and keeping the cheapest would void the guarantee, because the cheapest is the most likely to have been certified by chance. A ladder that stops below the full budget is risky even when it never stops early, so its always-continue rule is the first element of its sequence, and if that rule is rejected CAB falls back to the full budget. The full budget alone is always a candidate and needs no certificate: when nothing cheaper can be certified, for lack of calibration data or of a reliable controller, CAB is the standard estimator.

4. Experiments

4.1. Setup

Protocol. All data splits, rules and configurations below were pre-registered before the new data were downloaded or processed; the paper's claims rest on the *blind* test sets, each evaluated once after every selection had been written to disk. The official benchmarks, which earlier versions of this work had evaluated, are reported as *development* benchmarks for comparability. The protocol and its single deviation are in the code release (Supp. B).

Indoor data. The 75 3DMatch [32] training scenes are permuted with a fixed seed: 9 scenes form the blind test set, 13 provide 3,000 calibration pairs, and the rest 1,000 fitting pairs; pairs overlap by more than 10 %. The blind pairs are reported in two parts, above 30 % overlap (2,092 pairs,

3DMatch-like) and between 10 and 30 % (908, 3DLoMatch-like). Development benchmarks: the official 3DMatch (1,623) and 3DLoMatch [15] (1,781) test pairs, with the same fitting and calibration pairs.

Outdoor data. The 1,310 blind pairs come from the 49 KITTI [12] raw drives that belong to none of the odometry sequences 0–10, paired by FCGF's rule [7] with GPS/IMU poses refined by ICP; 22 pairs on which the refinement failed were dropped by a pre-registered rule. 300 odometry training pairs fit the controller and 1,238 other odometry training and validation pairs certify it; the 555 odometry test pairs are the development benchmark.

Correspondences and estimators. FPFH [25] descriptors (5 cm voxels indoors, 30 cm outdoors); for every source point the nearest target descriptor, sorted by Lowe ratio and capped at $N = 4{,}000$. FCGF [7] features are computed for every split with the official weights by our re-implementation of the network, which reproduces the released test features up to rounding (Supp. B). FCGF was trained on all 3DMatch training scenes and on odometry sequences 0–5, so it has no blind indoor test; it is fitted on the indoor fitting pairs, certified on the 1,331 pairs of the 3DMatch validation scenes and tested on the development benchmarks, and outdoors it is certified on 1,107 pairs of 42 raw drives and tested blind on the 203 pairs of the other 7. SC^2-PCR [6] and TurboReg [30] are re-implemented on CPU. At the full budget they come within 0.5–1.3 pp of the published recall on 3DMatch and 3DLoMatch with FCGF, and within 2.0–4.4 pp with FPFH (Supp. B). Budgets are $\{250, 500, 1000, 2000, 4000\}$.

Metrics. A pair is registered if RE $\leq 15^\circ$ and TE ≤ 30 cm indoors, and RE $\leq 5^\circ$ and TE ≤ 60 cm outdoors; recall (RR) is over all pairs. The *loss* of a configuration is the recall lost against the full budget with its one-sided 95 % upper bound (scene-stratified bootstrap [10]); it is *non-inferior* if the bound is below 1 pp (2 pp at low overlap), and its *risk* is the fraction of pairs registered by the full budget but not by CAB.

Cost. *Wall-clock speed-ups* are measured *live*: one process with the default multi-threaded BLAS on a laptop CPU (AMD Ryzen 7 5700U, 8 cores), the same 300 random test pairs per set for every configuration, and the order of the runs rotated from pair to pair. Each run includes everything the configuration needs, such as the ordering in CAB or the sampling signal in FastMAC. Because time depends on the machine, we also count *operations*: the dominant arithmetic of every stage a pair visits (compatibility tests, graph products, consensus and hypothesis scoring, $O(N^2)$ to $O(N^3)$ in the budget N) plus the $O(P^2 + Nm^2)$ of our ordering, averaged over the full test set (Supp. B). Live and offline stopping decisions agree on at least 97 % of the pairs in every live run.

4.2. Main results

Table 1 reports the blind test sets, with operating points chosen on the fitting pairs and certified on the calibration pairs before any blind pair was processed. On the held-out scenes, CAB runs SC^2-PCR 1.56× and TurboReg 1.49× faster than the standard full budget (1.69× and 1.72× fewer operations) and loses 0.05 and 0.00 pp of recall: the one-sided 95 % bounds, 0.19 and 0.10 pp, are below the 1 pp margin, so both are statistically non-inferior, and the realized risks of 0.10 and 0.05 % are inside the certified 1 %. At low overlap and on the KITTI raw drives, nearly every pair needs the full budget: CAB loses at most 0.08 pp, with every risk inside the level, but pays for the first stage it tries, and its live time is 0.94× and 0.95× (low overlap) and 0.93× and 0.91× (raw drives) of the full budget's speed; Sec. 4.5 explains why. The development benchmarks (Tab. 2) agree: 1.58× and 1.42× on 3DMatch, 1.08× and 1.07× on KITTI odometry, nothing saved on 3DLoMatch, and risks inside the level throughout. Filling the budgets by distinctiveness instead (ratio order) is much slower for SC^2-PCR on the held-out scenes (0.95×), whose small budgets succeed far less often (55.9 vs. 68.8 % at $N = 250$); for TurboReg the ratio order is faster (1.64×), because TurboReg's first stage becomes several times more expensive on consistent budgets (Sec. 4.6).

4.3. The guarantee under shift

A certificate is only worth having if it holds on data that differ from the calibration data. Figure 1(c) collects the realized risk on every test set: CAB stays within the 1 % level in all 40 cases (at most 0.55 %). Two alternatives do not, each for a different reason.

Marginal certificates hold only for the calibration mix of difficulties. On the blind KITTI raw drives, whose pairs are much harder than the odometry calibration pairs (61.8 vs. 92.3 % full-budget recall), the marginal certificate for TurboReg with the ratio order loses 1.37 % of the pairs, and with FCGF on 3DMatch and 3DLoMatch the marginal certificates lose up to 1.29 %; the group-conditional ones stay within the level in every case, as Proposition 1 predicts. Within one of the three difficulty groups, the marginal certificates' test risk reaches 1.67 %, against at most 0.67 % for CAB on the FPFH test sets (Supp. D). On KITTI odometry, a single threshold certified for CAB's own ladder loses 1.62 % of TurboReg's pairs, where the group-conditional thresholds lose 0.36 %.

Fixed shortcuts carry no certificate. Uniform sampling of 2,000 correspondences loses 3.35 and 3.82 % of the pairs registered by the full estimator on the held-out scenes, and 4.83 and 5.27 % on 3DLoMatch (Sec. 4.4).

Learned descriptors. With FCGF, the controller and its certificate are fitted on FCGF features, and the guarantee holds again (risks of at most 0.55 %), but the savings are

Table 1. **Blind test sets** (pre-registered, evaluated once; group-conditional certificates). RR: recall (%). Loss: recall lost against the full budget (pp) with its one-sided 95 % upper bound; ✓: non-inferior (bound below 1 pp; 2 pp at low overlap). Time: live wall-clock speed-up over the full budget on 300 random test pairs per set, all configurations timed on the same pairs. Ops: reduction of the hardware-independent operation count over the full test set (Sec. 4.1). Risk: pairs registered by the full budget but not by CAB (%, target ≤ 1). "Ratio order" fills the budgets with the most distinctive correspondences instead of the most consistent ones. With FCGF on the KITTI raw drives (203 pairs), CAB runs the full budget (Sec. 4.3).

	Held-out scenes (2,092 pairs)					Held-out scenes, low overlap (908)					KITTI raw drives (1,310)				
Method	RR	Loss [UB]	Time	Ops	Risk	RR	Loss [UB]	Time	Ops	Risk	RR	Loss [UB]	Time	Ops	Risk
FPFH correspondences															
SC²-PCR															
Full budget (standard)	85.2	–	1.00×	1.00×	–	40.5	–	1.00×	1.00×	–	61.8	–	1.00×	1.00×	–
CAB, ratio order	85.0	0.14 [0.33] ✓	0.95×	1.05×	0.19	40.3	0.22 [0.45] ✓	0.89×	1.01×	0.22	61.8	−0.08 [0.00] ✓	0.89×	1.18×	0.00
CAB (ours)	**85.1**	0.05 [0.19] ✓	**1.56**×	**1.69**×	0.10	**40.5**	0.00 [0.00] ✓	**0.94**×	**1.02**×	0.00	**61.7**	0.08 [0.23] ✓	**0.93**×	**1.14**×	0.08
TurboReg															
Full budget (standard)	85.7	–	1.00×	1.00×	–	40.2	–	1.00×	1.00×	–	62.6	–	1.00×	1.00×	–
CAB, ratio order	85.6	0.10 [0.29] ✓	1.64×	1.64×	0.19	40.1	0.11 [0.33] ✓	1.01×	1.03×	0.11	62.4	0.15 [0.38] ✓	1.00×	1.21×	0.15
CAB (ours)	**85.7**	0.00 [0.10] ✓	**1.49**×	**1.72**×	0.05	**40.2**	0.00 [0.00] ✓	**0.95**×	**1.05**×	0.00	**62.5**	0.08 [0.38] ✓	**0.91**×	**1.15**×	0.31

Table 2. **Development benchmarks** (official test sets, evaluated by earlier versions of this work; columns as in Tab. 1). FCGF block: controller fitted and certified on FCGF features of the fitting and calibration pairs. †CAB runs the full budget (Sec. 4.3).

	3DMatch (1,623 pairs)					3DLoMatch (1,781)					KITTI odometry (555)				
Method	RR	Loss [UB]	Time	Ops	Risk	RR	Loss [UB]	Time	Ops	Risk	RR	Loss [UB]	Time	Ops	Risk
FPFH correspondences															
SC²-PCR															
Full budget (standard)	81.5	–	1.00×	1.00×	–	36.6	–	1.00×	1.00×	–	92.3	–	1.00×	1.00×	–
CAB, ratio order	81.5	0.06 [0.18] ✓	0.93×	1.05×	0.06	36.6	0.06 [0.17] ✓	0.90×	1.00×	0.06	92.3	0.00 [0.00] ✓	1.18×	1.63×	0.00
CAB (ours)	**81.5**	0.00 [0.18] ✓	**1.58**×	**1.61**×	0.12	**36.6**	0.06 [0.17] ✓	**0.96**×	**1.01**×	0.06	**92.1**	0.18 [0.54] ✓	**1.08**×	**1.42**×	0.18
TurboReg															
Full budget (standard)	81.9	–	1.00×	1.00×	–	36.6	–	1.00×	1.00×	–	92.3	–	1.00×	1.00×	–
CAB, ratio order	81.6	0.31 [0.62] ✓	1.59×	1.52×	0.37	36.5	0.06 [0.22] ✓	1.03×	1.03×	0.11	92.1	0.18 [0.54] ✓	1.39×	1.78×	0.18
CAB (ours)	**81.7**	0.18 [0.37] ✓	**1.42**×	**1.58**×	0.18	**36.5**	0.06 [0.17] ✓	**0.97**×	**1.04**×	0.06	**92.1**	0.18 [0.72] ✓	**1.07**×	**1.41**×	0.36
FCGF correspondences															
SC²-PCR															
Full budget (standard)	92.5	–	1.00×	1.00×	–	58.0	–	1.00×	1.00×	–	93.5	–	1.00×	1.00×	–
CAB, ratio order	92.5	0.06 [0.37] ✓	1.00×	1.06×	0.31	57.7	0.34 [0.62] ✓	1.01×	1.10×	0.45	93.5	0.00 [0.00] ✓	1.00×†	1.00×	0.00
CAB (ours)	**92.2**	0.37 [0.68] ✓	**1.04**×	**1.07**×	0.43	**57.7**	0.28 [0.56] ✓	**1.08**×	**1.15**×	0.45	**93.5**	0.00 [0.00] ✓	**1.00**×†	**1.00**×	0.00
TurboReg															
Full budget (standard)	92.6	–	1.00×	1.00×	–	58.4	–	1.00×	1.00×	–	93.5	–	1.00×	1.00×	–
CAB, ratio order	92.6	0.00 [0.12] ✓	0.95×	1.03×	0.06	58.3	0.11 [0.28] ✓	0.99×	1.06×	0.11	93.5	0.00 [0.00] ✓	1.00×†	1.00×	0.00
CAB (ours)	**92.1**	0.49 [0.86] ✓	**1.01**×	**1.10**×	0.55	**58.2**	0.17 [0.51] ✓	**1.03**×	**1.19**×	0.39	**93.5**	0.00 [0.00] ✓	**1.00**×†	**1.00**×	0.00

small (Tab. 2): the pairs FCGF solves are solved by the first stage already, and its few failures are hard to tell from successes. On the KITTI raw drives the controller cannot tell them apart at all: its fitting pairs are FCGF's own training pairs, on which nearly every first stage succeeds, so it learns no failure signal; the certificate rejects every early stop on the calibration drives, and CAB runs the full budget, although 250 correspondences already register 92.1 % of the blind pairs. A certificate thus protects against a weak controller, at the price of speed; it cannot replace fitting data with failures.

4.4. Equal-time comparison with cheaper estimators

Is adaptivity needed, or would any cheaper way of running the estimator do as well? Figure 1 and Tab. 3 compare CAB with five alternatives on the same blind pairs and the same live clock. *Fewer seeds or cliques* barely save time: the cheapest settings still take 555 and 629 ms (full budget: 664 and 1,009 ms), because building the compatibility graph over all correspondences dominates. Neither does *FastMAC-style sampling* as published: its signal on the full second-order graph alone takes 573 ms on a CPU; on the first-order graph the signal costs 281 ms, and at CAB's time this variant registers 3.1 pp fewer pairs (SC²-PCR). *Fixed budgets* reach any time but trail CAB by 1.5–2.0 pp (SC²-PCR) and 2.4–2.5 pp (TurboReg) at its time, because a fixed rule must serve the hardest pairs it wants to keep, while CAB spends the extra correspondences only where the first budget failed. The strongest baseline is *uniform sampling*, and CAB beats it at equal or lower cost: with $N = 2{,}000$ random correspon-

Table 3. **Equal-time comparison** on the blind held-out scenes. Each cheaper variant is a recall–time curve closed by the full budget (live time, recall on every second test pair, 1,046 pairs). Min.: cheapest configuration (ms). At CAB: the variant's recall at CAB's time (%), and CAB's advantage on the same pairs (pp). Match: time the variant needs to reach CAB's recall.

Variant	Min.	At CAB	Δ	Match
SC²-PCR: full budget 664 ms, 86.3 %; CAB 427 ms, 86.3 %				
Fewer seeds (ratio 0.025, 0.01)	555	–	–	664 (1.6×)
Uniform sampling	26	85.2	+1.1	664 (1.6×)
FastMAC-style, 1st-order graph	304	83.2	+3.1	664 (1.6×)
FastMAC-style, 2nd-order graph	595	–	–	841 (2.0×)
Fixed budget, ratio order	27	84.8	+1.5	664 (1.6×)
Fixed budget, compat. order	28	84.3	+2.0	664 (1.6×)
TurboReg: full budget 1,009 ms, 87.6 %; CAB 676 ms, 87.6 %				
Fewer cliques (K_1=500, 250)	629	86.5	+1.3	1,009 (1.5×)
Uniform sampling	6	85.9	+1.7	1,009 (1.5×)
FastMAC-style, 1st-order graph	311	85.1	+2.5	1,009 (1.5×)
FastMAC-style, 2nd-order graph	612	86.6	+2.1	never
Fixed budget, ratio order	10	85.1	+2.5	1,009 (1.5×)
Fixed budget, compat. order	60	85.2	+2.4	1,009 (1.5×)

Table 4. **When CAB pays off.** Per test set, blind sets first (SC²-PCR / TurboReg): recall of the cheapest budget ($N = 250$, compatibility order, %), share of pairs no stage registers (%), live speed-up of CAB (†full budget, see Sec. 4.3) and realized risk (%).

Test set	N=250	Unsolvable	Speed-up	Risk
Held-out scenes	68.8 / 68.5	14 / 14	1.56 / 1.49	0.10 / 0.05
Held-out scenes, low overlap	23.5 / 23.6	58 / 58	0.94 / 0.95	0.00 / 0.00
KITTI raw	6.3 / 7.2	38 / 37	0.93 / 0.91	0.08 / 0.31
KITTI raw (FCGF)	92.1 / 92.1	7 / 8	1.00† / 1.00†	0.00 / 0.00
3DMatch	65.1 / 65.0	18 / 17	1.58 / 1.42	0.12 / 0.18
3DLoMatch	21.4 / 21.6	62 / 62	0.96 / 0.97	0.06 / 0.06
KITTI	12.8 / 14.2	8 / 7	1.08 / 1.07	0.18 / 0.36
3DMatch (FCGF)	86.8 / 87.0	6 / 7	1.04 / 1.01	0.43 / 0.55
3DLoMatch (FCGF)	44.1 / 43.6	40 / 40	1.08 / 1.03	0.45 / 0.39
KITTI (FCGF)	92.4 / 92.4	6 / 6	1.00† / 1.00†	0.00 / 0.00

dences SC²-PCR registers 84.5 % of the pairs in 286 ms against 86.3 % for CAB in 427 ms (McNemar $p = 0.01$), and TurboReg 84.4 against 87.6 % ($p = 1.1 \times 10^{-6}$). On 3DLoMatch (Supp. E) there is nothing to win at equal time: every cheaper variant that saves at least 5 % of the time loses at least 2.6 pp (SC²-PCR) and 1.3 pp (TurboReg) of recall, whereas CAB, whose certificate bounds the loss at 1 pp, correctly runs almost every pair to the full budget (1.06× and 1.06× the full time).

4.5. When it pays off, and why

When it pays off. Table 4 lines up all test sets. The speed-up is set mainly by the data, through how often a small budget already succeeds and how few pairs are unsolvable: with the same wrapper, it is largest where 250 correspondences register about two thirds of the pairs (held-out scenes and 3DMatch), small on KITTI odometry, where most pairs

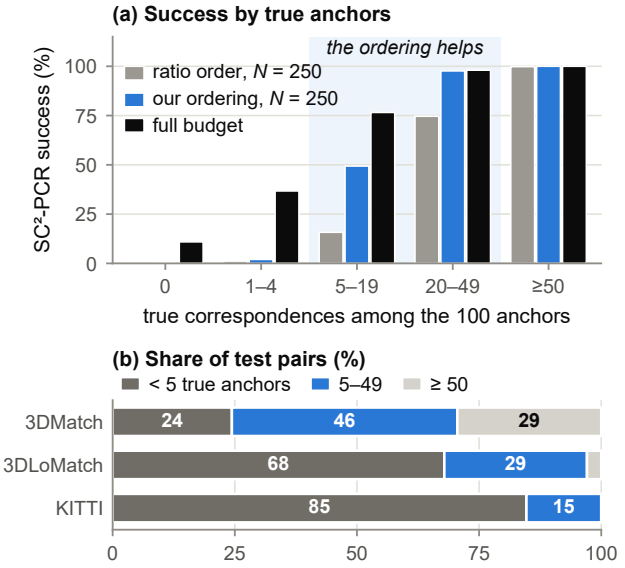


Figure 3. **Which pairs a cheap budget can rescue.** (a) SC²-PCR success on the 3DMatch and 3DLoMatch test pairs, by the number of true correspondences among the 100 anchors: at $N = 250$, our ordering (blue) recovers most of the full-budget success for 5–49 true anchors (shaded) and nothing below. (b) Share of the test pairs with fewer than 5, 5–49 and at least 50 true anchors: only the middle group (blue) can be rescued by a cheap budget, and its share differs most between test sets.

need 2,000 correspondences (12.8 % at $N = 250$), and it vanishes at low overlap and on the raw drives, where 58 and 38 % of the pairs are unsolvable and only 6.3 % of the raw-drive pairs are registered at $N = 250$. The risk stays within the level in every row: where there is nothing to save, CAB spends the full budget rather than losing pairs, at the price of the first stage it tries (lowest speed-up 0.91×, TurboReg on KITTI raw), which also lengthens the slowest pairs (Sec. 5). The next two paragraphs explain the two extremes of the data.

Middle-difficulty pairs. The ordering changes what the cheapest budget can do: with $N = 250$, SC²-PCR registers 65.1 % of 3DMatch pairs instead of 50.8 %, and 12.8 % of KITTI pairs instead of 1.6 %. Figure 3(a) shows for which pairs. On 3DMatch and 3DLoMatch, the gain is 23.0 to 33.6 pp for pairs with 5–49 true anchors and below 1 pp elsewhere: there, pairs with fewer than five are rarely registered even with the full budget, and pairs with fifty or more were solved already. An ordering reorganises the correspondences a pair has; it cannot create inliers. Figure 3(b) shows why the benefit varies across datasets: 68 % of 3DLoMatch pairs have fewer than five true anchors, against 24 % on 3DMatch. On KITTI 85 % do, yet the full budget registers 92.3 % of the pairs: their inliers are rarely among the most distinctive correspondences, from which the anchors are drawn, so the

Table 5. **Ordering ablation** on the 3DMatch validation scenes: inliers among the first 250 correspondences and success (%) at $N = 250/500/1000$. Full budget: 71.4 % (SC^2-PCR) and 72.2 % (TurboReg).

		SC^2-PCR			TurboReg		
Ordering	Inl.@250	250	500	1k	250	500	1k
Lowe ratio	27.8	51.3	61.2	67.5	51.7	61.2	67.2
One pass	82.6	52.8	57.8	63.8	52.9	58.2	63.8
Two passes (ours)	104.7	58.8	62.6	66.8	59.4	62.9	66.6

pairs are solvable but only with large budgets, which is why the speed-up on KITTI is modest (Tab. 4).

Unsolvable pairs. Supp. Tab. 7 bounds what any stopping rule can achieve on the same ladder. Without failure detection, the best possible speed-up for SC^2-PCR on the held-out scenes is 2.92× with the compatibility order (2.50× with the ratio order): the ordering raises the ceiling itself, and CAB reaches 79 % of it with marginal and 57 % with group-conditional certificates. The remaining headroom of the full oracle comes from *unsolvable* pairs, which are 14 % of the held-out pairs but 58 % at low overlap and 59–60 % of its full-budget time. Recognising them early is hard: after the first stage, the value-of-continuing model predicts final success with an AUROC of 0.82 on 3DMatch but 0.68 on 3DLoMatch. On low-overlap data the bottleneck is therefore not stopping too late on solvable pairs, but spending the full budget on pairs that no budget can register.

4.6. Ablations

Anchors. Table 5 isolates the choice of anchors on the validation scenes. One pass, with the most distinctive correspondences as anchors, triples the inliers in the cheap budget (82.6 vs. 27.8) but changes SC^2-PCR's success at $N = 250$ by only +1.5 pp and *lowers* it at 500 and 1,000 (−3.4 and −3.7 pp); two passes gain +7.5 and +1.4 pp at $N = 250$ and 500, the budgets at which the selected ladders start. Mean inlier yield is a misleading design target: it is dominated by easy pairs, while success is decided by the hard ones.

Certificates. With marginal certificates CAB is faster on the held-out scenes (2.31× and 2.03× offline, against 1.67× and 1.57×), at risks of 0.33 and 0.38 % there, but these certificates exceed their level elsewhere (Sec. 4.3). Conditioning on difficulty trades some speed for a guarantee that survives this shift (Proposition 1). Varying the risk level and the number of groups (Supp. D) trades speed for risk smoothly: from $\varepsilon = 1$ to 5 %, SC^2-PCR's speed-up on 3DMatch rises from 1.55× to 2.25× with the realized risk always below ε, and four or more groups need more calibration pairs than our pools contain.

Estimator cost depends on its input. On consistency-ordered budgets, SC^2-PCR's first stage becomes cheaper (52 → 34 ms; fewer seeds survive non-maximum suppression), whereas TurboReg's becomes 4.3× more expensive at an unchanged operation count, because a consistent subgraph fills all of its candidate cliques. This is why the ordering buys TurboReg accuracy more than time, and why wall-clock time stays our primary cost measure [9].

5. Limitations and conclusion

Limitations. Our estimators are CPU re-implementations; on a GPU the cost gap between small and large budgets narrows, and the speed-ups must be measured again. CAB lowers the mean time per pair, not the worst case: a pair that needs the whole ladder also pays for the stages tried before, up to 36 % more than one full run at the 99th percentile (Supp. C), and where nearly every pair needs the full budget, CAB is slightly slower than the full budget on average (Tab. 1). The development benchmarks were evaluated by earlier versions of this work, and what they revealed (the failure of marginal certificates) shaped the design; the blind test sets were not, and confirm its results. A certificate needs calibration data, and its speed needs fitting data with failures: on random calibration subsets, the group-conditional certificate appeared only from about a thousand pairs (Supp. D), and with FCGF on KITTI, whose fitting pairs are FCGF's own training pairs, CAB can only run the full budget. FCGF has seen every 3DMatch training scene, so learned descriptors are evaluated blind only outdoors.

Conclusion. Not every pair needs every correspondence, but a cheaper estimator is only useful if one knows what it loses. Three lessons follow from our experiments. First, filling small budgets with mutually consistent correspondences and stopping adaptively makes graph-based estimators markedly faster where the data allow, at statistically non-inferior recall, on scenes never used during development. Second, the guarantee must be conditional on difficulty and must cover the configuration actually used: marginal certificates exceed their level when the mix of difficulties changes and a fixed shortcut loses pairs without warning, while difficulty-conditional certificates of a candidate chosen beforehand held on every blind and development test set. Third, a certificate is only as fast as its controller: fitted on the descriptor in use and on data with failures, it pays off; without failures to learn from, the certificate keeps the full budget. Where pairs are mostly unsolvable, as at low overlap, the time goes to pairs that no budget can register: predicting registrability before searching, rather than stopping better while searching, is the next step.

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